

CSCE 679 DATA VISUALIZATION · MODULE 04 · LECTURE NOTES

Colour

Where most published figures go wrong.

MODULE OUTCOMES

- Explain the three scale types and when each applies.
- Explain why the rainbow scale misleads.
- Explain perceptual uniformity and luminance ordering.
- Explain the accessibility requirements.
- Choose and check a colour scale.

1 Three kinds of scale

Scale	What it is for	How it is built
Categorical	Nominal data, with no order implied.	Distinguishable hues at similar luminance — see §3.
Sequential	Ordered data running from a low value to a high one.	Monotonic luminance, with one or two hues — see the callout.
Diverging	Data with a meaningful midpoint.	Two sequential ramps meeting at a neutral colour at the midpoint.

A diverging scale asserts that the midpoint means something — a zero, a baseline, a long-run average, a threshold — so using one on data with no natural midpoint invents a reference that the reader will believe, and the neutral colour will be read as 'normal'. The invented-midpoint error is common and easy to state, and it is one of the few colour errors that changes the conclusion rather than merely the legibility.

LUMINANCE CARRIES THE ORDERING, WHICH IS WHY SEQUENTIAL SCALES MUST BE MONOTONIC IN IT

Hue is not ordered (Module 02 §1's rank seven) — **there is no perceptual sense in which green is 'between' blue and yellow**, whatever the wavelengths do, and readers cannot reliably order hues even when told the intended order.

But luminance is ordered, and reliably so — darker and lighter are unambiguous — **which means a sequential scale must increase monotonically in luminance**, and **the hue is then decoration** that helps with discrimination but carries no order.

And that gives the test: convert the scale to greyscale, and it should run smoothly from dark to light — **which is a five-second check** and is **CSCE 671 Module 02 §2's greyscale rule** applied specifically to a colour scale.

Which immediately condemns the rainbow (§2), because **its luminance goes up and down several times across the range** — bright at yellow, dark at both ends — **and therefore encodes no consistent order at all.**

2 The rainbow problem

1. NON-MONOTONIC LUMINANCE

it is bright at yellow and dark at both blue and red, so the apparent ordering reverses mid-scale and the brightest region is in the middle.

2. PERCEPTUALLY NON-UNIFORM

equal steps in the data are unequal steps in perceived difference. The green region is a wide band of near-identical colour; the cyan-to-blue transition is a sharp perceptual edge.

3. IT CREATES FALSE BOUNDARIES

those sharp transitions read as contours in the data. Readers see structure that is in the colour map and not in the measurement.

4. AND IT FAILS FOR COLOUR VISION DEFICIENCY

the red-green axis carries much of the range.

SO: the rainbow scale invents boundaries and hides real gradients, which is the worst possible combination for a quantitative encoding.

It invents boundaries and hides real gradients — which is the worst possible combination for a quantitative encoding, and is why this is a defect rather than a preference. The false-boundary point is the one that persuades scientists: a reader examining a rainbow-coloured field will identify features at the cyan-blue transition that have no counterpart in the data, and will fail to see a real gradient that falls inside the green band. That is a measurement being misreported by its presentation.

AND THE REPLACEMENTS ARE DESIGNED TO FIX EXACTLY THOSE FOUR THINGS

The viridis family and its relatives are constructed to be monotonic in luminance and approximately uniform in perceived difference — so **equal steps in the data look equally different to a reader**, which is the property the encoding requires.

Which means no false boundaries, because **there are no sharp perceptual transitions to be mistaken for contours in the data** — the second and third defects are addressed by the same construction.

And they are designed to remain ordered under the common colour vision deficiencies, because the luminance ramp survives even when the hue discrimination does not (CSCE 632 §3) — which is a direct consequence of the monotonic luminance rather than a separate feature.

So the switch is free and strictly better — the scales ship with every plotting library — **which makes continuing to use a rainbow scale a decision rather than a default**, and **one worth flagging in review** rather than tolerating.

3 Categorical colour

- **Keep the luminance similar across the categories**, because **a luminance difference implies an order** (§1's callout) **that nominal data does not have** — and readers will take the darkest category to be the most or the largest.
- **Use at most five to seven colours** (Module 02 §2's capacity) — **and if you have more categories, group them or facet rather than adding more colours**, which turns the channel into a legend lookup.
- **Reserve grey for the unimportant** — which is **the most effective single technique in categorical colour**: colour the one or two series that matter and make all the others grey, which uses the preattentive pop-out of CSCE 671 Module 02 §3 deliberately.
- **Respect existing conventions where they exist**, and **beware the ones that are cultural** — **red for loss and green for gain is not universal**, and neither is the political colour mapping, which inverts between countries.
- **And label the series directly rather than using a legend** wherever you can, since **a legend is a memory task** (CSCE 671 Module 03 §2) that has to be performed for every mark. **Direct labelling beats a legend** because it removes the lookup, **and it is almost always possible for a small number of series**.

4 Checking

- **The greyscale test. Convert the figure to greyscale: a sequential scale must still be ordered, and a categorical one must still be distinguishable or must carry a second channel** (shape, pattern, direct label).
- **The colour-vision simulation**, for the common deficiencies — **which takes seconds with any of several free tools and catches the red-green encoding** immediately (CSCE 632 §3).
- **And the print test**, since figures are still printed, photocopied, and projected badly — **which is the greyscale test with worse contrast** and catches insufficient luminance separation.
- **Plus the standing rule: never use colour as the only channel** — add shape, pattern, position, or a direct label, so that the figure degrades gracefully rather than becoming unreadable.
- **Which together take under a minute and catch most of the colour errors in published work** — **which makes omitting them quite hard to defend**, and is why this module ends on them rather than on the theory.

AND THE HONEST SUMMARY OF THIS MODULE

Colour is the channel people reach for first and the one with the most rules — because it is perceptually complicated and because it is the easiest thing in a figure to change, so it absorbs attention disproportionately.

The rules reduce to three: match the scale type to the data type (§1), make the ordering carried by luminance (§1's callout), and never let colour be the only channel (§4).

And the checks reduce to one: convert to greyscale and see whether it still works — which subsumes most of the others, since it tests the luminance ordering, the colour-only encoding, and the print case simultaneously.

Which is a small enough set to apply every single time — and the rainbow scale persists in published science precisely because nobody applied it, not because anybody defended it.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Wilke, the colour chapters (free)	The whole module — the scale types, the pitfalls, and the examples, free in full.
Borland & Taylor — Rainbow Color Map (Still) Considered Harmful (free)	§2's four defects , with the figures — short, and the false-boundary demonstration is the persuasive part.
Colorbrewer, and the viridis documentation (free)	§1 and §3 as working tools — with the colour-blind-safe and print-safe filters, which do §4's work for you.
Crameri, Shephard & Heron — The misuse of colour in science (free)	§2 measured across a literature — how prevalent the problem is, and what it has cost.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Classify five figures' colour scales** as categorical, sequential, or diverging, and say whether each matches its data.
- 2 Find a diverging scale on data with no midpoint**, and say what it implies.
- 3 Convert a rainbow scale to greyscale** and plot its luminance against position.
- 4 Render the same field in rainbow and in viridis**, and ask five people to describe the features.

- 5 **Count the features they report that are not in the data.**
- 6 **Build a categorical palette with varying luminance** and ask people which category is largest.
- 7 **Grey out all but one series** in a multi-series chart, and compare.
- 8 **Replace a legend with direct labels** and time a lookup from each.
- 9 **Run all three checks** on five of your own figures.
- 10 **Find a cultural colour convention** that inverts between two regions.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name the three scale types and what each is for.
- 2 What does a diverging scale assert, and what is the error?
- 3 Why does luminance carry the ordering and hue not?
- 4 Give the five-second test for a sequential scale.
- 5 Give the four defects of the rainbow scale.
- 6 Which defect persuades scientists, and why?
- 7 What do the perceptually uniform scales fix, and how does that help colour vision deficiency?
- 8 Give five rules for categorical colour, and the most effective technique.
- 9 Give the three checks and the standing rule.
- 10 Reduce this module to three rules and one check.